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Common ownership and corporate violations: Evidence from China

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ABSTRACT

Using a sample of Chinese listed firms from 2007 to 2022, we provide robust evidence of a negative relationship between common ownership and corporate violations, and document that the potential mechanisms are improved corporate governance and enhanced information transparency. Our cross-sectional analyses reveal that the negative effect of common ownership on corporate violations is more pronounced among firms with a higher level of media monitoring and those with a lower proportion of state ownership. By distinguishing between different types of common ownership, we identify that common state ownership and common foreign ownership serve as key forces in reducing corporate violations. Overall, our findings are consistent with the notion that common ownership serves as an effective form of "synergistic governance" and plays a critical role in curbing corporate violations.

1. Introduction

Common ownership of public firms occurs when an investor cross-holds equity stakes in peer firms within the same industry (Ramalingegowda et al., 2021; Chen et al., 2023a). Common ownership has become a prevalent phenomenon, raising widespread concern in both developed and emerging markets (Azar et al., 2018; Athira and Lukose, 2023). In China, the number of common owners and the proportion of firms with common owners are increasing year on year (Fig. 1). Although some literature suggests that common ownership can result in governance synergies among co-owned firms (e.g., Kang et al., 2018; He et al., 2019), others document that it can cause anti-competitive effects across an industry and weaken oversight over management (e.g., He and Huang, 2017; Lu et al., 2022; Antón et al., 2023). Given the persistent controversy over the governance implications of common ownership, we focus on corporate violations, which are a direct manifestation of governance failures, to assess the impact of common ownership.

Corporate violations are an important consideration for investors, as they can affect shareholder wealth, firm value, and the overall health of the capital market (Fich and Shivdasani, 2007; Chen et al., 2023b). Prior research documents that external law environment (La Porta et al., 2002; Kedia and Rajgopal, 2011), industry and geographic regulations (Dyck et al., 2010; Kedia and Rajgopal, 2011; Wang, 2013), auditors (Dyck et al., 2010), and the media (Miller, 2006; Dyck et al., 2008, 2010; Heese et al., 2022), play central roles in monitoring corporate violations. Internally, firm-specific characteristics, such as performance pressure and managers' self-interested behaviors (Bell and Carcello, 2000; Agrawal and Chadha, 2005; Khanna et al., 2015), as well as board and CEO characteristics (Chen et al., 2006; Efendi et al., 2007; Khanna et al., 2015), can also influence corporate violations. Yet, despite an extensive body of research examining various determinants of corporate violations, the potential impact of common ownership on firms'

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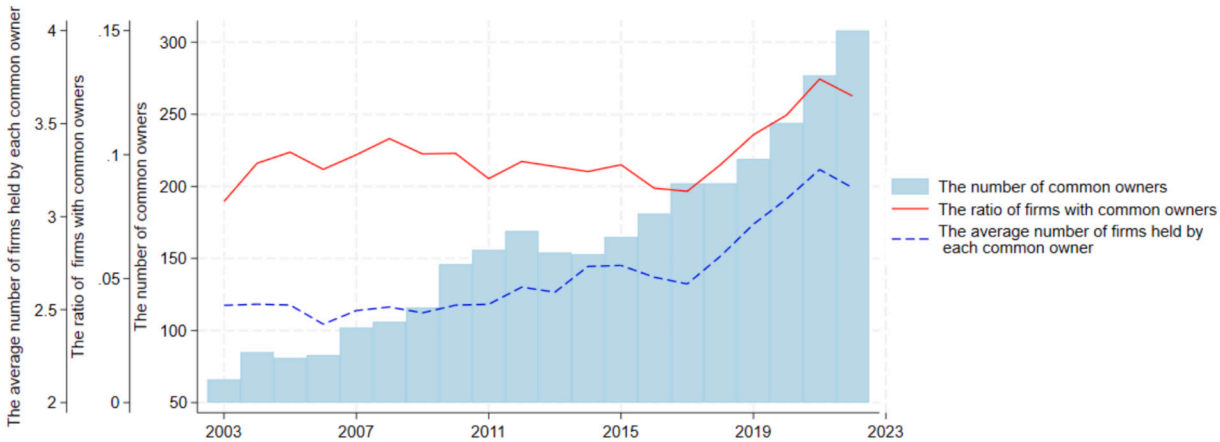

Fig. 1. Common ownership in Chinese stock market.
The figure plots the number of common owners, the ratio of firms with common owners to all firms, and the average number of firms held by each common owner for each year from 2003 to 2022. The bar chart represents the number of common owners in the market, the solid line stands for the ratio of firms with common owners to all firms, and the dashed line indicates the average number of firms held by each common owner.

violation behaviors has received surprisingly little attention. Our study extends prior research on corporate violations by introducing the effect of common ownership.

We predict that common ownership will be a proactive force curbing corporate violations. This prediction is grounded in the theoretical frameworks of the positive effect of common ownership on corporate governance and information transparency. First, common owners serve as effective monitors. They can directly intervene in corporate governance through voice, voting, and exit threats to prevent managers from pursuing private benefits (Kang et al., 2018; Edmans et al., 2019; He et al., 2019). They also have access to more knowledge, expertise, and experience across their portfolio, which enhances their monitoring capabilities (Kang et al., 2018). Additionally, they benefit from economies of scale, particularly in terms of information acquisition and monitoring multiple firms in the same industry, which further enhances their monitoring efficiency (Chen et al., 2023a). These advantages work to reduce managerial opportunism, thus curbing the occurrence of corporate violations. Second, common ownership tends to enhance a firm's transparency. Co-owned firms are not only motivated to voluntarily disclose more information (Park et al., 2019; Pawliczek et al., 2022), but they are also subject to greater external oversight (Edmans et al., 2019; Jang et al., 2022). This enhanced transparency increases the likelihood of managers' wrongdoings being detected, which helps to suppress corporate violations.

To investigate whether common ownership affects corporate violations, we empirically examine the relationship using a sample of Chinese A-share listed firms over the period 2007 to 2022. Our baseline regression results suggest a negative relationship between common ownership and corporate violations. The effect is not only statistically significant but also economically relevant. To address potential endogeneity issues, we employ a series of identification strategies, including propensity score matching (PSM), Heckman two-stage, instrumental variable, placebo, and DiD tests. The consistency of results across these different methods confirms the reliability of our baseline findings and strengthens the validity of our causal inference. Our results remain robust after adopting alternative independent variables, sample period, model, and fixed effects. Moreover, we identify improvements in corporate governance and information transparency as potential mechanisms through which common ownership affects corporate violations. In cross-sectional analyses, we find that the negative relationship between common ownership and corporate violations is more pronounced for firms with a greater extent of external media monitoring and those with a relatively lower proportion of state-owned shares. Additional analyses indicate that, among various types of common ownership, common state ownership and common foreign ownership exhibit the most significant negative relationship with corporate violations.

Our study makes three main contributions. First, this paper adds to the literature on the effect of common ownership. While some research suggests that common ownership creates governance synergies across portfolio firms, referred to as the "synergistic governance" effect (e.g., Kang et al., 2018; Ramalingegowda et al., 2021), other studies suggest that common ownership decreases competition and fosters coordination or tacit collusion among firms in the industry, referred to as the "collusion" effect (e.g., He and Huang, 2017; Azar et al., 2018). There is also a third strand of literature that suggests common ownership has only a limited impact (e.g., Lewellen and Lowry, 2021). Our study contributes to this literature by documenting a tendency toward synergistic governance as a result of common ownership. Unlike prior studies on governance synergies that examine the role of common ownership in specific aspects of managerial misconduct, such as curbing tax avoidance (Athira and Lukose, 2023), constraining insider trading (Dai et al., 2024), mitigating earnings management (Ramalingegowda et al., 2021), and deterring bad news hoarding (Chen et al., 2023a), we focus on corporate violations, a direct and comprehensive manifestation of governance failure, to assess the impact of common ownership. Furthermore, we provide evidence that improved corporate governance and increased information transparency serve as mechanisms through which common ownership acts as an effective form of "synergistic governance".

Second, we enrich the literature on the determinants that influence corporate violations. An existing body of literature documents some of these factors, including external forces such as the legal and regulatory environment, the media, and auditors (La Porta et al.,

2002; Kedia and Rajgopal, 2011; Dyck et al., 2010; Heese et al., 2022), as well as the internal characteristics of the boardroom and CEO (Chen et al., 2006; Efendi et al., 2007; Khanna et al., 2015). We complement this line of literature by introducing a cross-firm factor, common ownership, and document that corporate violations decrease in portfolio firms of common owners. Our study highlights that common owners serve as one of the governance forces that deter the occurrence and enhance the detection of corporate violations. Given the widespread prevalence of common ownership, our novel evidence on its impact on corporate violations deserves attention for academic research and regulatory policy.

Third, our paper provides nuanced insights into the differential impact of various types of common ownership on corporate violations. By distinguishing between common institutional ownership, common enterprise ownership, and common individual ownership, we find that common enterprise ownership is the key force in curbing corporate violations. Further decomposing common enterprise ownership into state-owned, non-state-owned, and foreign-owned categories demonstrates that common state ownership and common foreign ownership are particularly effective at reducing corporate violations. These findings indicate that the identity and nature of common owners matter significantly, not just the presence of common ownership itself. Thus, this study should be of interest to researchers, regulators, and corporations, as it provides evidence that the governance efficiency of common ownership differs between different types of common ownership.

The remainder of our paper is organized as follows: Section 2 provides literature review and hypothesis. Section 3 describes the sample and research design. Section 4 presents empirical results. Section 5 concludes.

2. Literature review and hypothesis development

2.1. Prior research on common ownership

Common owners simultaneously hold shares in competing firms within the same industry, shifting their goals from maximizing the value of individual firms to maximizing returns across an entire portfolio (e.g., Azar et al., 2018; He et al., 2019). This can generate countervailing effects: efficiency gains from coordinated governance (the "synergistic governance" effect) versus anti-competitive and collusion effects through reduced market discipline (the "collusion" effect). One strand of the literature suggests that common ownership creates governance synergies due to more information advantages and greater corporate governance (e.g., Kang et al., 2018; Edmans et al., 2019; He et al., 2019). However, an opposing strand of literature points out that common ownership can facilitate market coordination or tacit collusion and decrease competition among firms in the common owner's portfolio (e.g., He and Huang, 2017; Lu et al., 2022; Antón et al., 2023), which creates a hidden social cost for the entire industry. There is also a third strand of literature suggesting that common ownership only has a modest impact on competition (Lewellen and Lowry, 2021; Lewellen and Lewellen, 2022). In fact, Lewellen and Lowry (2021) document that any anti-competitive effects observed likely stem from other factors, such as firms' or industries' differential responses to the 2008 financial crisis. They find little robust evidence that common ownership affects firm behavior when using modified empirical approaches that are less sensitive to these issues.

2.2. Hypothesis development

Corporate violations refer to actions taken by firms that violate the law or deviate from social norms, resulting in a loss of value for the firms and harming the interests of investors (e.g., Holzman et al., 2021; Chen et al., 2023b). Violations are typically driven by managers' opportunistic motives to maximize their own interests, which often arise from inadequate internal governance systems (Chen et al., 2006; Donelson et al., 2017). Building on the literature that documents the governance synergies effect of common ownership, we propose that common ownership will reduce corporate violations due to improved corporate governance and increased information transparency.

The first channel through which common ownership reduces corporate violations is improved corporate governance. Common ownership can improve corporate governance for three reasons. First, common owners can directly intervene in corporate governance via mechanisms such as voice and exit, thereby preventing managers from pursuing their own private benefits. For example, they are more likely to vote against management on shareholder-sponsored governance proposals, and their exit is effective because a reduction in their holdings in a specific firm signals a pessimistic outlook to the market (He et al., 2019; Edmans et al., 2019). Additionally, they can force poorly performing executives to turnover and vote on director candidates to monitor managerial behavior and curb opportunistic insider trading (Kang et al., 2018; Dai et al., 2024). Such interventions can prompt managers to take actions that better align with the interests of shareholders.

Second, owning multiple blockholdings within a portfolio not only strengthens the incentives for common owners to monitor the firms in their portfolio but also improves their governance capabilities. Common owners have a strong incentive to maximize the value of their portfolio, which drives them to leverage their advantages to enhance monitoring efficiency (He et al., 2019; Ramalingegowda et al., 2021). They can also spread knowledge and governance expertise among co-owned peer firms, acting as an information bridge among portfolio firms and thereby reducing monitoring costs (Kang et al., 2018; Ramalingegowda et al., 2021; Athira and Lukose, 2023).

Third, common owners benefit from economies of scale, which also helps to strengthen their monitoring efficiency. For example, when an industry is influenced by unfortunate macroeconomic factors, a common owner does not need to develop a different coping strategy for each firm in their portfolio. Rather, one good strategy should suffice for all firms. Moreover, common owners generally have access to overarching information and expertise, making them better equipped to identify misconduct and bad news (Ramalingegowda et al., 2021; Chen et al., 2023a; Liu and He, 2024).

Given that common ownership can constrain managerial opportunistic behaviors and mitigate agency conflicts, managers will be more cautious and compliant in making decisions. Several studies provide evidence that stronger external monitoring and internal governance result in fewer violations. For example, Dyck et al. (2010), Karpoff and Lou (2010), and Wang et al. (2010) find that auditors, media, industry regulators, employees, short sellers, and underwriters can play a monitoring role that prevents firms from engaging in misconduct and violations. Therefore, we expect that the monitoring role of common ownership can suppress the opportunistic behaviors of managers, which in turn inhibits corporate violations.

The second channel through which common ownership reduces corporate violations is enhanced information transparency. Common owners can improve the transparency of portfolio firms for at least two reasons. First, co-owned firms are more motivated to disclose information. Common owners aim to promote cooperation and resource sharing. Firms with common owners can have greater voluntary disclosure because common ownership reduces proprietary cost concerns of disclosure and internalizes the externality benefits of disclosure (Park et al., 2019). The sharing of information among portfolio firms, coupled with the enhanced monitoring capabilities of common owners, also contributes to greater transparency in the information disclosed by these firms.

Second, co-owned firms can attract greater public attention and are subject to stricter oversight. Under external pressure, these firms become more transparent, which deters them from manipulating earnings and concealing bad news (Ramalingegowda et al., 2021; Chen et al., 2023a). Additionally, common owners can systematically compare financial reporting and other information among industry peers, enabling them to identify anomalies and potential misconduct more effectively than isolated investors. A firm's violation behaviors have negative externalities (He et al., 2019; Park et al., 2019). When a violation occurs, false disclosures and misleading statements may distort the investment decisions of peer firms. Therefore, common ownership can reduce the prevalence of false information in financial reports and help avoid losses for portfolio firms (He et al., 2019).

Common ownership can improve information transparency, which increases the probability that misconduct will be detected and thereby deters such behaviors. Jin and Myers (2006) and Bleck and Liu (2007) find that managers opt to support bad projects to maximize their benefits in markets with significant information asymmetry. Once violations are discovered, the perpetrators must bear the consequences, which include returning illegally obtained profits and facing significant penalties. Given that common ownership can increase a firm's information transparency, we expect it to help suppress corporate violations.

Based on the above discussion, we propose that common ownership can improve corporate governance and enhance information transparency, thereby leading to a decrease in corporate violations. Therefore, we propose our hypothesis as follows:

Hypothesis. Common ownership is associated with lower corporate violations.

While our primary hypothesis predicts a negative relationship between common ownership and corporate violations, we acknowledge compelling theoretical arguments supporting an alternative perspective. The "collusion" effect of common ownership may increase corporate violations. First, common ownership may diminish product market competition (Hansen and Lott, 1996; He and Huang, 2017; Lu et al., 2022), thereby weakening an important external governance mechanism (Hart, 1983; Chhaochharia et al., 2017). In less competitive environments, managers may have greater latitude to engage in opportunistic behaviors that serve their own private interests rather than shareholders' interests. Second, common ownership might facilitate information sharing that enables sophisticated violations across portfolio firms through complex related-party transactions (Brooks et al., 2018). These countervailing effects create a tension that ultimately renders the relationship between common ownership and regulatory violations an empirical question worthy of rigorous investigation.

3. Research design

3.1. Data and sample

Our initial sample consists of Chinese A-share listed firms over the period of 2007–2021. The data are primarily collected from the China Stock Market & Accounting Research (CSMAR) database and supplemented by the Chinese Research Data Services (CNRDS) dataset and the DIB database. We extract common ownership holdings information from the CSMAR database and define a holding as a block if it exceeds 5 % of the firm's outstanding shares. Given the revision of the Chinese Accounting Standards in 2007, this study selects 2007 as the initial research year. We exclude the following observations from the initial sample: (1) financial firms; (2) ST and *ST firms; (3) firms with leverage ratios exceeding 1; and (4) firms with missing data for key variables. After these exclusions, we obtain 37,957 firm-year observations. To mitigate the impact of extreme values, all continuous variables are winsorized at the 1st and 99th percentiles.

3.2. Variable construction

3.2.1. Common ownership

To measure the impact of common ownership in a given fiscal year, following He and Huang (2017) and Athira and Lukose (2023), we focus on blockholders owning at least 5 % of a firm's outstanding shares and construct two measures: (1) *Cross5_Dum* is a dummy variable that equals 1 if the focal firm and at least one of its industry peers are simultaneously held by at least one blockholder in any quarter of the fiscal year, and 0 otherwise. (2) *Cross5_Num* is measured as the natural logarithm of 1 plus the average number of common blockholders that hold the focal firm and its industry peers across the four quarters of a fiscal year. We use the 3-digit industry code to identify industry peers based on the Guidelines for the Industry Classification of Listed Companies issued by the China Securities Regulatory Commission in 2012.

Table 1
Summary statistics.

Variable	N	Mean	SD	Min	p25	p50	p75	Max
VioDummy	37,957	0.178	0.383	0.000	0.000	0.000	0.000	1.000
VioNum	37,957	0.305	0.808	0.000	0.000	0.000	0.000	10.000
Cross5_Dum	37,957	0.104	0.305	0.000	0.000	0.000	0.000	1.000
Cross5_Num	37,957	0.068	0.210	0.000	0.000	0.000	0.000	1.609
Size	37,957	22.043	1.283	19.719	21.106	21.854	22.772	26.069
Lev	37,957	0.423	0.206	0.051	0.258	0.417	0.579	0.884
ROA	37,957	0.039	0.060	−0.239	0.015	0.039	0.069	0.198
OCF	37,957	0.048	0.071	−0.163	0.008	0.047	0.089	0.248
Growth	37,957	0.183	0.399	−0.554	−0.012	0.120	0.286	2.499
Invest	37,957	0.052	0.049	0.000	0.016	0.038	0.073	0.236
Top1	37,957	0.351	0.147	0.094	0.235	0.330	0.451	0.746
Indirect	37,957	0.373	0.053	0.300	0.333	0.333	0.429	0.571
Dual	37,957	0.277	0.447	0.000	0.000	0.000	1.000	1.000
SOE	37,957	0.373	0.484	0.000	0.000	0.000	1.000	1.000
Fix	37,957	0.219	0.164	0.002	0.092	0.185	0.311	0.710
Age	37,957	16.814	5.965	4.000	13.000	17.000	21.000	32.000
BIG4	37,957	0.032	0.177	0.000	0.000	0.000	0.000	1.000

This table presents descriptive statistics of our key variables. Definitions of all variables are detailed in Appendix Table 1.

3.2.2. Corporate violations

Drawing on the methods of Zhang (2018) and Chen et al. (2023b), the corporate violations are proxied by *VioDummy* and *VioNum*. *VioDummy* is a dummy variable that equals 1 when a firm is sanctioned for violations by the regulatory authorities, and 0 otherwise. *VioNum* is calculated as the sum of the number of violations committed by the firm.

3.2.3. Control variables

Following Zhang (2018), Jiang et al. (2022), and Chen et al. (2023b), we include a series of firm-level financial characteristics and external government variables to control for the potential drivers of corporate violations. These control variables include firm size (*Size*), leverage (*Lev*), profitability (*ROA*), operating cash flow (*OCF*), sales growth (*Growth*), capital expenditure (*Invest*), the largest shareholding (*Top1*), independent directors (*Indirect*), CEO duality (*Dual*), state ownership (*SOE*), fixed assets (*Fix*), firm age (*Age*), and whether the firm is audited by a Big Four accounting firm (*BIG4*). For detailed definitions of the variables, please refer to Appendix Table 1.

3.3. Regression specifications

To examine the effect of common ownership on corporate violations, we construct the following baseline regression model:

$$Vio_{i,t+1} = \beta_0 + \beta_1 Cross_{i,t} + \beta_n Controls_{i,t} + Firm\ FE + Year\ FE + \varepsilon_{i,t} \tag{1}$$

where *i* represents the firm, and *t* represents the year. The dependent variable is corporate violations (*Vio*) as proxied by *VioDummy* and *VioNum.* The independent variable is common ownership (*Cross*), proxied by *Cross5_Dum* and *Cross5_Num*. In our regression specification, we include firm fixed effects to absorb time-invariant firm characteristics. We also include year fixed effects to mitigate concerns that our results might be driven by time trends. Standard errors are clustered at the firm level, and $\varepsilon_{i,t}$ represents the residual term. The coefficient for *Cross*, $\beta 1$, captures the relationship between common ownership and corporate violations. Specifically, we employ ordinary least squares (OLS) to estimate the coefficients in Eq. (1).

3.4. Summary statistics

Table 1 presents the summary statistics for the key variables. There is significant variation in firms' violations: the mean value of *VioDummy* is 0.178, indicating that approximately 17.8 % of firm-years in our sample involve violations. The average *VioNum* is 0.305. In terms of common ownership measures, the mean value of *Cross5_Dum* is 0.104, which means that 10.4 % of the sample firms have common ownership. The mean value of *Cross5_Num* is 0.068. The standard deviation of *Cross5_Dum* (*Cross5_Num*) is 0.305 (0.210). The sample means of control variables are similar to those reported in previous studies (Chen et al., 2023b). Appendix Table 2[1] reports the Pearson correlation coefficients for the main variables. The results show that the common ownership measures (*Cross5_Dum* and *Cross5_Num*) are negatively correlated with the corporate violation measures (*VioDummy* and *VioNum*), providing preliminary support for our hypothesis.

[1] Appendix Tables 2–5 are included in the online appendix due to space limitations.

Table 2
The effect of common ownership on corporate violations.

	(1)	(2)	(3)	(4)
	VioDummy	*VioNum*	*VioDummy*	*VioNum*
Cross5_Dum	−0.030***	−0.083***		
	(−2.91)	(−4.22)		
Cross5_Num			−0.051***	−0.139***
			(−3.24)	(−4.64)
Size	0.046***	0.135***	0.046***	0.135***
	(6.33)	(7.82)	(6.33)	(7.82)
Lev	0.045	0.115	0.045	0.114
	(1.55)	(1.59)	(1.55)	(1.59)
ROA	−0.595***	−1.374***	−0.594***	−1.373***
	(−10.76)	(−10.17)	(−10.75)	(−10.16)
OCF	−0.047	−0.112	−0.048	−0.113
	(−1.30)	(−1.34)	(−1.31)	(−1.36)
Growth	−0.003	0.003	−0.003	0.003
	(−0.56)	(0.26)	(−0.56)	(0.26)
Invest	−0.064	−0.201	−0.064	−0.201
	(−1.08)	(−1.60)	(−1.07)	(−1.60)
Top1	−0.111**	−0.342***	−0.112**	−0.343***
	(−2.34)	(−2.97)	(−2.35)	(−2.98)
Indirect	−0.030	−0.096	−0.030	−0.096
	(−0.40)	(−0.63)	(−0.40)	(−0.63)
Dual	−0.002	0.010	−0.002	0.010
	(−0.17)	(0.45)	(−0.17)	(0.45)
SOE	−0.098***	−0.250***	−0.098***	−0.250***
	(−4.23)	(−4.63)	(−4.22)	(−4.62)
Fix	−0.114***	−0.365***	−0.114***	−0.364***
	(−3.33)	(−4.57)	(−3.32)	(−4.56)
Age	−0.008	−0.023	−0.008	−0.023
	(−0.70)	(−1.09)	(−0.70)	(−1.08)
BIG4	−0.043***	−0.125***	−0.043***	−0.124***
	(−2.82)	(−4.14)	(−2.79)	(−4.10)
Constant	−0.577**	−1.931***	−0.576**	−1.928***
	(−2.30)	(−3.76)	(−2.30)	(−3.76)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	37,957	37,957	37,957	37,957
Adjusted R^2	0.250	0.299	0.250	0.299

This table presents the baseline regression results. The dependent variables are *VioDummy* and *VioNum*. The key explanatory variables are *Cross5_Dum* and *Cross5_Num*. We include firm and year fixed effects in all columns. Definitions of all variables are detailed in Appendix Table 1. All continuous variables are winsorized at the 1st and 99th percentiles. T-statistics with standard errors clustered at the firm level are reported in parentheses. *, **, and *** denote *P*-values of 10 %, 5 %, and 1 %, respectively.

4. Empirical results

4.1. Baseline results

Table 2 presents our baseline results based on the regression specified in Eq. (1). The dependent variables are *VioDummy* in Columns (1) and (3) and *VioNum* in Columns (2) and (4). The independent variables are *Cross5_Dum* in Columns (1)–(2), and *Cross5_Num* in Columns (3)–(4). The results show that both coefficients of *Cross5_Dum* and *Cross5_Num* are negative and statistically significant at the 1 % level. The economic magnitude of the effect is also substantial. As suggested in Column (2), compared to those in non-co-owned firms, the number of violations in co-owned firms decreases by 27.2 % relative to the sample mean. In Column (4), a one-standard-deviation increase in common ownership results in a 9.57 % decrease in corporate violations relative to the sample mean. These results lend support to the hypothesis that common ownership can reduce corporate violations.

4.2. Endogeneity tests

Our baseline regression results may have potential endogeneity issues in terms of identifying the relationship between common ownership and corporate violations. To address concerns associated with self-selection bias, omitted variables, and reverse causality, we conduct a series of tests using the following approaches: propensity score matching, Heckman analysis, instrumental variable approach, placebo test, and DiD analysis.

4.2.1. Propensity score matching analysis

To eliminate the possibility that fundamental differences between co-owned firms and non-co-owned firms may be driving our

Table 3
Endogeneity tests.

Panel A: Propensity score matching results

	(1)	(2)	(3)	(4)
	VioDummy	*VioDummy*	*VioNum*	*VioNum*
Cross5_Dum	−0.030*		−0.066**	
	(−1.82)		(−2.25)	
Cross5_Num		−0.051**		−0.099**
		(−2.20)		(−2.42)
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	7018	7018	7018	7018
Adjusted R^2	0.304	0.304	0.374	0.375

Panel B: Heckman two-stage results

	the first stage	the second stage			
	(1)	(2)	(3)	(4)	(5)
	Cross5_Dum	*VioDummy*	*VioNum*	*VioDummy*	*VioNum*
CSI300in	0.246**				
	(2.03)				
Cross5_Dum		−0.030***	−0.085***		
		(−2.80)	(−4.17)		
Cross5_Num				−0.048***	−0.139***
				(−2.98)	(−4.47)
IMR		−0.014	−0.037	−0.014	−0.037
		(−1.23)	(−1.37)	(−1.23)	(−1.38)
Controls	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	No	No	No	No
Firm FE	No	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
N	36,746	36,746	36,746	36,746	36,746
Adjusted R^2/Pseudo R^2	0.194	0.253	0.302	0.253	0.302

Panel C: Instrumental variable results

	the first stage		the second stage			
	(1)	(2)	(3)	(4)	(5)	(6)
	Cross5_Dum	*Cross5_Num*	*VioDummy*	*VioNum*	*VioDummy*	*VioNum*
Dividend	0.011**	0.009***				
	(2.55)	(2.89)				
Cross5_Dum			−3.282**	−8.257**		
			(−2.31)	(−2.37)		
Cross5_Num					−4.329**	−10.893***
					(−2.56)	(−2.64)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	37,516	37,516	37,516	37,516	37,516	37,516
Cragg-Donald Wald F statistic	12.895	17.749	NA	NA	NA	NA

Panel D: DiD results

	(1)	(2)
	VioDummy	*VioNum*
Treat×Post	−0.054***	−0.168***
	(−3.97)	(−6.13)
Controls	Yes	Yes
Firm FE	Yes	Yes
Year FE	Yes	Yes
N	32,892	32,892
Adjusted R^2	0.249	0.301

This table presents the results of the endogeneity tests. Panel A presents the results of the one-to-one PSM procedure. Panel B presents the results of the Heckman two-stage analysis. Panel C presents the results of the instrumental variable analysis. Panel D presents the DiD results. Definitions of all variables are detailed in Appendix Table 1. All continuous variables are winsorized at the 1st and 99th percentiles. T-statistics with standard errors clustered at the firm level are reported in parentheses. *, **, and *** denote P-values of 10 %, 5 %, and 1 %, respectively.

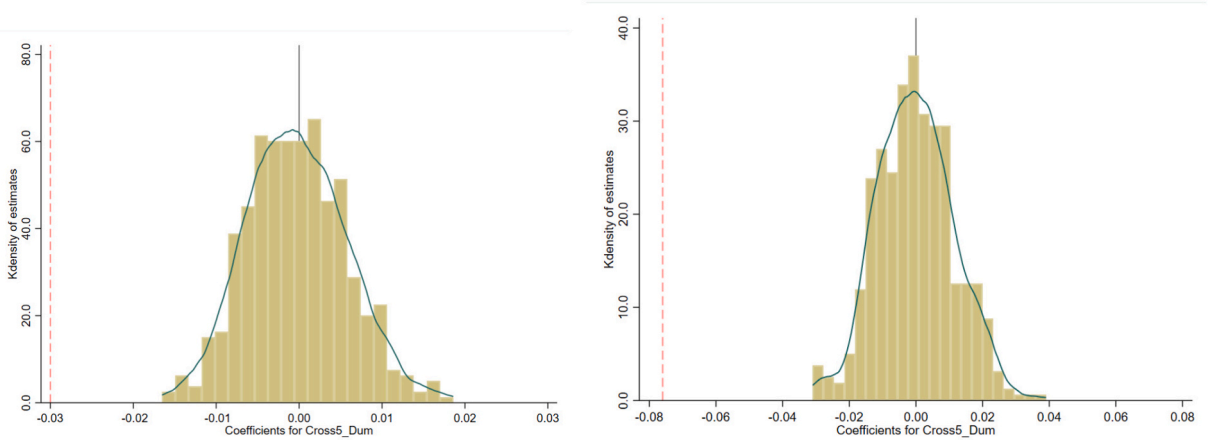

Fig. 2. Placebo test (coefficient).
This figure plots the empirical distribution of the estimated coefficients from a placebo test based on 1000 simulations using the false *Cross5_Dum*. The left side of Fig. 2 shows the distribution of the coefficient estimates when the dependent variable is *VioDummy*. The right side of Fig. 2 shows the distribution of the coefficient estimates when the dependent variable is *VioNum*. The vertical dotted lines (from left to right) show the results of Columns (1) and (2) in Table 2, respectively.

findings, we perform a propensity score matching analysis to address this potential selection bias. First, we estimate a probit model to predict the likelihood of common ownership presence. The covariates are all the control variables. Then, following the literature (Hsu et al., 2023; Yao et al., 2024), we utilize a one-to-one PSM procedure to select a matched control group. Specifically, we match each observation with common ownership to one control firm in the same industry in the same year. Through this procedure, we obtain 3509 samples for the treatment group and 3509 samples for the control group. For interested readers, Appendix Table 3 provides the results of the covariate balance checks on the mean difference between the treatment and the matched control groups. We find that the differences after matching are statistically insignificant, indicating no significant differences between the treatment group and the matched control group. Finally, we re-estimate Eq. (1) using the matched sample. The results are presented in Panel A of Table 3. The coefficients of Cross5_*Dum* and *Cross5_Num* remain negative and significant, verifying that our findings are robust.

4.2.2. Heckman analysis

There is another concern regarding self-selection caused by omitted variables that could result in a biased coefficient—namely, that investors may prefer investing in certain types of firms. Firms in the same industry may share the same characteristics, such as similar market trends or technological levels. These common characteristics may lead to common ownership linkages. In other words, our results may be driven by a firm's industry-specific factors rather than by common ownership. Following the prior procedure (Heckman, 1979), we conduct a Heckman analysis to mitigate concerns about investors' self-selection. Lennox et al. (2012) propose the use of restriction variables to estimate the omitted variables in the regression analyses. Our paper imposes exclusion restrictions to implement the Heckman procedure. In the first stage, we use probit modeling to estimate the inverse Mills ratios (*IMR*). Following the literature (Yao et al., 2024), we use *CSI300in* as the exclusion restriction variable. This is because firms newly included in the Shanghai Shenzhen 300 Stock Market Index (CSI 300 index) tend to receive increased attention from investors. *CSI300in* is a dummy variable, which equals 1 if a firm is newly added to the CSI 300 index in that year, and 0 otherwise. In Panel B of Table 3, we report the first-stage estimation results, which show that *CSI300in* is positively and significantly associated with common ownership. In the second stage, we re-estimate the regression of the effect of common ownership on corporate violations, adding the *IMR* variable as a control variable. In Columns (2)–(5), the coefficients of *IMR* are insignificant, suggesting that the potential endogeneity from self-selection is weak. The coefficients of *Cross5_Dum* and *Cross5_Num* are negative and significant. So, our results are robust after controlling for unobserved factors.

4.2.3. Instrumental variable analysis

Our findings might be attributed to the fact that firms with fewer corporate violations attract common owners to invest. That is, our findings may be affected by the issue of reverse causality. Thus, we adopt an instrumental variable approach to further mitigate potential endogeneity concerns. Previous literature has found that investors generally have specific stock preferences, such as those paying high dividends, stocks in firms listed on indexes, and stocks in large firms (Ferreira and Matos, 2008; Aggarwal et al., 2011). Since corporate dividends are directly related to investment returns for shareholders, most will expect the firm to maintain consistent

and stable dividend payments. Aggarwal et al. (2011) and Athira and Lukose (2023) use the dividend payments as an instrumental variable for institutional ownership. Paying higher dividends may imply that the firm has strong cash flow and profitability, but this does not directly lead to fewer corporate violations. This is because dividend payments do not directly affect a firm's regulatory environment, violation costs, managers' business strategies, or their risk appetite. Thus, dividend payments are unlikely to directly affect corporate violations. As a result, we use dividend payment (*Dividend*) as an instrumental variable for common ownership. *Dividend* is a dummy variable that equals 1 if the firm pays a cash dividend in that year, and 0 otherwise. Panel C of Table 3 reports the results, with Columns (1) and (2) presenting the results for the first stage of the regressions. As expected, *Dividend* is positively and significantly associated with common ownership, consistent with common owners preferring to invest in firms with higher dividend payouts. Columns (3)–(6) present the second-stage results. The coefficients on *Cross5_Dum* and *Cross5_Num* are all negative and statistically significant at the 1 % level, confirming the reliability of our baseline findings. To assess instrument strength, we conduct a weak identification test. Following the empirical rules proposed by Cragg and Donald (1993) and Stock and Yogo (2005), we calculate the Cragg-Donald Wald F statistic. Columns (1) and (2) of Panel C show that the Cragg-Donald Wald F statistics are all above 10, confirming the validity of *Dividend* as an instrumental variable.

4.2.4. Placebo test

The observed negative relationship between common ownership and corporate violations may be influenced by unobserved factors. To assess whether our findings are driven purely by chance, we conduct a placebo test following established methods (Chetty et al., 2009; Ferrara et al., 2012). More specifically, we generate a random treatment group from our sample and construct a "false" common ownership variable. To increase the identification power, we repeat this exercise 1000 times. We then re-estimate the baseline model Eq. (1) using the "false" common ownership. The distribution of estimated coefficients of the placebo test is provided in Fig. 2. The left side of Fig. 2 plots the coefficient distribution of "false" *Cross5_Dum* on *VioDummy*. The right side of Fig. 2 plots the coefficient distribution of "false" *Cross5_Dum* on *VioNum*. The vertical dotted lines (from left to right) show the results from Columns (1) and (2) of Table 2, respectively. These lie outside the range of the estimated false coefficients, indicating that there are no unobserved factors driving the baseline regression results.

4.2.5. DiD analysis

To further address endogeneity concerns, we examine changes in common ownership over time, focusing specifically on firms transitioning from having no common ownership to now having common ownership. We employ the following DiD model to estimate the differences in corporate violations associated with these changes in common ownership:

$$Vio_{i,t+1} = \beta_0 + \beta_1 Treat \times Post_{i,t} + \beta_n Controls_{i,t} + Firm\ FE + Year\ FE + \varepsilon_{i,t} \tag{2}$$

where *Treat* is a dummy variable that equals 1 if the firms have changed from having no common ownership to having common ownership, and 0 otherwise.[2] *Post* is a dummy variable that equals 1 for years after the firm has acquired common ownership, and 0 otherwise. In our sample, the majority of firms have experienced a change in common ownership from none to some. We exclude observations with frequent changes in common ownership. If the coefficient on the interaction term (*Treat*×*Post*) is significantly negative, it means that the change in common ownership can impede firms' violation behaviors. The interaction term (*Treat*×*Post*) captures the effect of the change in common ownership on firms' violation behaviors. Panel D of Table 3 reports the results of the DiD analysis. As shown in Columns (1) and (2), the coefficients of *Treat*×*Post* are all negative and significant, supporting that an increase in common ownership could lead to a reduction in corporate violations.

4.3. Robustness tests

4.3.1. Alternative measures of common ownership

To further test the robustness of our results, we replace our explanatory variable with two alternative measures: (1) *Density* and (2) *Cross5_Share*. Prior studies (Azar et al., 2018; Lewellen and Lowry, 2021) use the traditional Herfindahl-Hirschman Index to measure common ownership concentration. Following Lewellen and Lowry (2021), we use this method to calculate the density of the ownership network, which is defined as the sum of the products of shareholding ratios. Specifically, *Density* is defined as:

$$Density_{i,t} = \sum_{k=1}^{K} \sum_{j=1}^{N} \omega_{k,t} \times \mu_{j,i,t} \times \mu_{j,k,t} \tag{3}$$

where $\omega_{k,t}$ represents the weight of each rival firm k in year t. We use the ratio of a firm's sales to total industry sales as the weight. $\mu_{j,i,t}$ represents investor j's ownership percentages in firm i during the year t. $\mu_{j,k,t}$ represents investor j's ownership percentages in firm k during the year t. As shown in Columns (1) and (2) of Panel A of Appendix Table 5, the coefficients for *Density* are significantly

[2] Treatment firms are firms that experience a change in common ownership from 0 to 1 only once during our sample period. Control firms are firms that have never had common ownership. Following Dai et al. (2024), if we redefine treatment firms as firms that have only once experienced a change in common ownership from 0 to 1 or from 1 to 0, the coefficient on *Treat*×*Post* remains significantly negative. The results are presented in **Appendix Table 4** for interested readers.

Table 4
Mechanism tests.

Panel A: Corporate governance

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	CODirector	*ICQ*	*SGA*	*ORR*	*CODirector*	*ICQ*	*SGA*	*ORR*
Cross5_Dum	0.118***	6.885**	−0.009***	−0.002**				
	(20.80)	(1.97)	(−3.15)	(−2.29)				
Cross5_Num					0.205***	13.560**	−0.016***	−0.003**
					(22.72)	(2.56)	(−3.51)	(−2.25)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	37,951	34,608	37,957	37,957	37,951	34,608	37,957	37,957
Adjusted R^2	0.707	0.348	0.737	0.415	0.728	0.348	0.737	0.415

Panel B: Information transparency

	(1)	(2)	(3)	(4)
	REM	*ASY*	*REM*	*ASY*
Cross5_Dum	−0.010**	−0.073***		
	(−2.01)	(−6.45)		
Cross5_Num			−0.019**	−0.099***
			(−2.43)	(−5.22)
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	37,957	37,740	37,957	37,740
Adjusted R^2	0.354	0.702	0.355	0.702

This table presents the results for the two potential mechanisms. Panel A presents the results about improved corporate governance mechanism. The dependent variables are proxies for corporate internal governance. Panel B presents the results about enhanced information transparency mechanism. The dependent variables are proxies for corporate information transparency. The key explanatory variables are *Cross5_Dum* and *Cross5_Num*. We include firm and year fixed effects in all columns. Definitions of all variables are detailed in Appendix Table 1. All continuous variables are winsorized at the 1st and 99th percentiles. T-statistics with standard errors clustered at the firm level are reported in parentheses. *, **, and *** denote P-values of 10 %, 5 %, and 1 %, respectively.

negative. Next, we use the shareholdings of common ownership (*Cross5_Share*) to measure the common ownership. *Cross5_Share* is measured by the average shareholdings of common ownership that holds the focal firm and industry peers across the four quarters of a fiscal year. The results are presented in Columns (3) and (4) of Panel A of Appendix Table 5. The results show that the coefficients for *Cross5_Share* are significantly negative at the 1 % level. The results suggest that our findings remain unchanged using different measures of common ownership.

4.3.2. Alternative thresholds of ownership proportions

In our baseline analysis, we construct common ownership using a 5 % threshold for block holders. One potential concern is that investors with higher shareholdings may have stronger incentives to engage in corporate governance. When an investor's shareholding is relatively low, the effect on governance may diminish. To address this concern, we redefine common owners as those who own at least 3 % or 1 % of the firm's outstanding shares. Then, we construct *Cross3_Dum* and *Cross1_Dum* to re-estimate Eq. (1). All other settings remain unchanged. As shown in Panel B of Appendix Table 5, the coefficients of *Cross3_Dum* and *Cross1_Dum* are negative and statistically significant. The results show that our findings remain unchanged after using different thresholds.

4.3.3. Alternative sample period

Another possible concern about our baseline findings is that the sample period overlaps with COVID-19. The ongoing disruption and uncertainty caused by the COVID-19 pandemic have dealt a severe blow to the global economy, as noted by Shehzad et al. (2020), Wu et al. (2023), and Lin et al. (2024). To prevent the pandemic's spread, China implemented a zero-Covid policy till December 2022. Similarly, Li et al. (2021) and Banerjee et al. (2023) show that the COVID-19 pandemic has led to disruptions in supply chains and substantial declines in corporate operational performance. Under pressure from the operating environment, managers might commit fraud to achieve their goals. However, the restrictions posed by the pandemic would make it difficult for regulators to detect violations. As a result, our findings are likely to be influenced by COVID-19. To eliminate this issue, we exclude the COVID-19 years and re-estimate our Eq. (1) using the 2007–2019 sample period. As shown in Panel C of Appendix Table 5, the conclusions remain negative and significant after addressing this concern.

4.3.4. Alternative model and fixed effects

In addition, we repeat our baseline regression with an alternative model and fixed effects. Given that our dependent variable,

Table 5
The impact of external monitoring.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Media = 1	Media = 0	Media = 1	Media = 0	Media = 1	Media = 0	Media = 1	Media = 0
	VioDummy	*VioDummy*	*VioNum*	*VioNum*	*VioDummy*	*VioDummy*	*VioNum*	*VioNum*
Cross5_Dum	−0.036***	0.003	−0.069***	−0.029				
	(−2.74)	(0.20)	(−3.11)	(−0.86)				
Cross5_Num					−0.054***	0.001	−0.114***	−0.044
					(−2.79)	(0.04)	(−3.54)	(−0.81)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Difference P-value	0.000***		0.000***		0.000***		0.000***	
N	17,238	18,451	17,238	18,451	17,238	18,451	17,238	18,451
Adjusted R^2	0.243	0.241	0.273	0.271	0.243	0.241	0.273	0.271

This table presents the cross-sectional analysis results concerning the impact of external monitoring. Columns (1), (3), (5), and (7) present the results of subsamples with a greater extent of external media monitoring. Columns (2), (4), (6), and (8) present the results of subsamples with a lower extent of external media monitoring. We include firm and year fixed effects in all columns. Definitions of all variables are detailed in Appendix Table 1. All continuous variables are winsorized at the 1st and 99th percentiles. T-statistics with standard errors clustered at the firm level are reported in parentheses. *, **, and *** denote *P*-values of 10 %, 5 %, and 1 %, respectively.

VioDummy, is a dummy variable, we re-estimate the baseline specification using a probit model. The results are presented in Columns (1) and (3) of Panel D of Appendix Table 5. Given that our dependent variable, *VioNum*, is a count variable, we re-estimate the baseline using a Poisson model. The results are presented in Columns (2) and (4) of Panel D of Appendix Table 5. In Panel E of Appendix Table 5, we include *Industry*×*Year* fixed effects to control for time-varying changes in industry differences. The coefficients of *Cross5_Dum* and *Cross5_Num* are negative and statistically significant, suggesting that the baseline regression results are robust.

4.4. Mechanism analyses

4.4.1. Corporate governance

According to our hypothesis, common ownership plays a role in corporate governance by leveraging its monitoring capabilities (Kang et al., 2018; Edmans et al., 2019; Ramalingegowda et al., 2021; Athira and Lukose, 2023). As corporate governance improves, management's financial decisions become more cautious and compliant, which in turn reduces corporate violations. To test whether corporate governance serves as a channel through which common ownership curbs corporate violations, we examine the effect of common ownership on corporate governance.

Specifically, we construct four measures to capture corporate governance following prior literature (Dai et al., 2024; Cheng et al., 2024; Singh and Davidson III, 2003; Jiang et al., 2010). The first measure, common ownership directors (*CODirector*), is defined as the ratio of directors appointed by common owners to total number of directors. The second measure, the quality of internal control (*ICQ*), is defined as the internal control index constructed by the DIB database. The third measure, SG&A expense (*SGA*), is defined as the amount of selling, general, and administrative expenses scaled by total sales. The fourth measure, other receivables (*ORR*), is defined as the amount of other receivables scaled by total assets.

The regression results are presented in Panel A of Table 4. In Columns (1) and (5), where the dependent variable is *CODirector*, the coefficients of *Cross5_Dum* and *Cross5_Num* are positive and significant. These results indicate that common owners gain board representation in their portfolio firms, allowing them to actively participate in corporate governance. In Columns (2) and (6), where the dependent variable is *ICQ*, the coefficients of *Cross5_Dum and Cross5_Num* are also positive and significant, suggesting that common ownership enhances the quality of internal controls for firms. In Columns (3) and (7), with *SGA* as the dependent variable, and in Columns (4) and (8), where the dependent variable is *ORR*, the coefficients of *Cross5_Dum and Cross5_Num* are both negative and significant, suggesting that common ownership can reduce corporate agency problems.

Collectively, these results provide evidence that common ownership improves corporate governance. Given that an effective internal governance system plays a crucial role in detecting and deterring corporate fraud and misconduct (e.g., Chen et al., 2006; Shan, 2013), our findings suggest that improved corporate governance serves as a mediator in the effect of common ownership on curbing corporate violations.

4.4.2. Information transparency

Common owners of multiple firms within the same industry can obtain supplementary information and achieve economies of scale in information acquisition (Chen et al., 2023a). They can also share their knowledge and experiences among co-owned peer firms, acting as information bridges within their portfolios, which enhances the information transparency of these firms. This generally translates to a greater likelihood of detecting and preventing corporate violations. To test whether information transparency serves as a channel through which common ownership curbs corporate violations, we examine the effect of common ownership on information transparency.

Specifically, we measure information transparency using two proxies. The first is real earnings management (*REM*), which is

Table 6
The impact of state ownership.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	SOE_Share = 1	SOE_Share = 0	SOE_Share = 1	SOE_Share = 0	SOE_Share = 1	SOE_Share = 0	SOE_Share = 1	SOE_Share = 0
	VioDummy	*VioDummy*	*VioNum*	*VioNum*	*VioDummy*	*VioDummy*	*VioNum*	*VioNum*
Cross5_Dum	−0.016	−0.026**	−0.047	−0.072***				
	(−0.68)	(−2.11)	(−1.00)	(−3.23)				
Cross5_Num					−0.020	−0.052***	−0.076	−0.133***
					(−0.57)	(−2.71)	(−1.00)	(−3.92)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Difference P-value	0.021**		0.003***		0.000***		0.000***	
N	7863	30,094	7863	30,094	7863	30,094	7863	30,094
Adjusted R^2	0.275	0.263	0.295	0.313	0.275	0.263	0.295	0.313

This table presents the cross-sectional analysis results concerning the impact of state ownership. Columns (1), (3), (5), and (7) present the results of subsamples with a higher proportion of state ownership. Columns (2), (4), (6), and (8) present the results of subsamples with a lower proportion of state ownership. We include firm and year fixed effects in all columns. Definitions of all variables are detailed in Appendix Table 1. All continuous variables are winsorized at the 1st and 99th percentiles. T-statistics with standard errors clustered at the firm level are reported in parentheses. *, **, and *** denote P-values of 10 %, 5 %, and 1 %, respectively.

constructed using the methodology developed by Roychowdhury (2006). The second is information asymmetry index (*ASY*), which is constructed following Amihud et al. (1997), Amihud (2002), and Pástor and Stambaugh (2003). In both measures, a higher value indicates lower information transparency. Using *REM* and *ASY* as dependent variables, the regression results are presented in Panel B of Table 4. As shown in Columns (1) to (4), the coefficients for *Cross5_Dum and Cross5_Num* are negative and significant. These results suggest that common ownership improves corporate transparency. Given that more transparent information makes it easier to detect misbehavior (Dyck et al., 2010; Xu et al., 2024), our results indicate that increased information transparency acts as a mediator, enabling common shareholders to deter management from engaging in violation activities.

4.5. Cross-sectional analyses

4.5.1. The impacts of media monitoring

Prior literature suggests that the media plays an external governance role in monitoring corporate violations (Dyck et al., 2010; Jiang et al., 2022). To examine whether this effect is complementary or substitutive, we investigate whether and how media monitoring changes the negative relationship between common ownership and corporate violations.

We use the negative media attention to measure media monitoring (*Media*) following Jiang et al. (2022). By classifying the observations above and below the year-industry median, we divided the sample into high and low media monitoring subsamples. We then re-estimate our baseline regressions using each subsample and present the results in Table 5. As shown, the coefficients of both *Cross5_Dum* and *Cross5_Num* are significantly negative for the high media monitoring subsample but are insignificant for the low media monitoring subsample. The difference between the coefficients in the two subsamples is significant. These results show that the negative effect of common ownership on corporate violations is more pronounced for firms with greater media monitoring, which implies that media monitoring and common ownership are complementary in deterring corporate violations.

4.5.2. The impacts of state ownership

According to the literature (e.g., Wang et al., 2004; Liu et al., 2024), state-owned firms are subject to direct government regulation and supervision. They also face greater public scrutiny and social pressure. State ownership can prompt firms to adopt stricter corporate governance structures and also make them more responsive to the public interests, which helps to supervise and control corporate behavior. In this section, we examine whether and how the deterrence effect of common ownership on violations varies with state ownership.

We measure state ownership (*SOE_Share*) as the proportion of a firm owned by the state following Hou et al. (2012). We classify the observations above and below the year-industry median into high and low state ownership subsamples. We then re-estimate Eq. (1) for both subsamples and present the results in Table 6. The results show that the coefficients of *Cross5_Dum* and *Cross5_Num* are significantly negative for the low state ownership subsample, and insignificant for the high subsample. The difference between the coefficients is significant, suggesting that the negative relationship between common ownership and corporate violations is more pronounced for firms with low state ownership.

4.5.3. The impacts of different types of common ownership

Compared to previous studies that primarily focus on common institutional ownership (e.g., Azar et al., 2018; Lewellen and Lowry, 2021), our sample from China provides an opportunity to distinguish the effects of different types of common ownership. To examine

Table 7
The impact of types of common ownership.

Panel A: The type of common ownership

	(1)	(2)	(3)	(4)
	VioDummy	*VioNum*	*VioDummy*	*VioNum*
CrossInst_Dum	−0.018	0.053		
	(−0.53)	(0.63)		
CrossEnt_Dum	−0.029***	−0.088***		
	(−2.72)	(−4.42)		
CrossInd_Dum	0.160	0.082		
	(1.39)	(0.62)		
CrossInst_Num			−0.029	0.073
			(−0.59)	(0.61)
CrossEnt_Num			−0.041***	−0.124***
			(−2.90)	(−4.65)
CrossInd_Num			0.231	0.117
			(1.39)	(0.62)
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	37,957	37,957	37,957	37,957
Adjusted R^2	0.250	0.299	0.250	0.299

Panel B: The type of common enterprise ownership

	(1)	(2)	(3)	(4)
	VioDummy	*VioNum*	*VioDummy*	*VioNum*
CrossSOE_Dum	−0.028**	−0.076***		
	(−2.08)	(−2.87)		
CrossNSOE_Dum	−0.016	−0.066		
	(−0.62)	(−1.54)		
CrossFOR_Dum	−0.040***	−0.116***		
	(−2.80)	(−5.19)		
CrossSOE_Num			−0.040**	−0.110***
			(−2.21)	(−3.03)
CrossNSOE_Num			−0.021	−0.093
			(−0.60)	(−1.55)
CrossFOR_Num			−0.055***	−0.164***
			(−2.72)	(−5.20)
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	37,957	37,957	37,957	37,957
Adjusted R^2	0.250	0.299	0.250	0.299

This table reports the regression results for the relation between common ownership of different types of investors and corporate violations. Panel A reports the results for common institutional owners, common enterprise owners, and common individual owners. Panel B reports the results for common state owners, common non-state owners, and common foreign owners. We include firm and year fixed effects in all columns. Definitions of all variables are detailed in Appendix Table 1. All continuous variables are winsorized at the 1st and 99th percentiles. T-statistics with standard errors clustered at the firm level are reported in parentheses. *, **, and *** denote P-values of 10 %, 5 %, and 1 %, respectively.

the impact of different types of common owners on corporate violations, we categorize common ownership based on the nature of the shareholders and construct several variables accordingly. The first is common institutional ownership (*CrossInst_Dum*), which equals 1 if an institutional investor simultaneously held shares in the focus firm and at least one of its industry peers in any quarter of the year, and 0 otherwise. The second is common enterprise ownership (*CrossEnt_Dum*), which equals 1 if an enterprise simultaneously held shares in the focus firm and at least one of its industry peers in any quarter of the year, and 0 otherwise. The third is common individual ownership (*CrossInd_Dum*), which equals 1 if an individual investor simultaneously held shares in the focus firm and at least one of its peers in any quarter of the year, and 0 otherwise. We also construct corresponding continuous variables (*CrossInst_Num, CrossEnt_Num*, and *CrossInd_Num*) by taking the natural logarithm of 1 plus the average number of each types of common ownership. Using these variables as explanatory variables, the regression results are presented in Panel A of Table 7. The coefficient of *CrossEnt_Dum* (*CrossEnt_Num*) is significantly negative, while the coefficients of *CrossInst_Dum* (*CrossInst_Num*) and *CrossInd_Dum* (*CrossInd_Num*) are insignificant. These results indicate that common enterprise ownership is the key force in curbing corporate violations.

To further explore the effect of common enterprise ownership, we break it down into three subtypes based on the nature of the common enterprise: common state ownership (*CrossSOE_Dum*), common non-state ownership (*CrossNSOE_Dum*), and common foreign ownership (*CrossFOR_Dum*). The three variables are measured similarly to those above, with a value of 1 indicating that common

ownership types are state-owned, non-state-owned, and foreign-owned enterprises, respectively. We also construct corresponding continuous variables (*CrossSOE_Num*, *CrossNSOE_Num*, and *CrossFOR_Num*) by taking the natural logarithm of 1 plus the average number of each subtypes of common enterprise ownership. As shown in Panel B of Table 7,[3] the coefficients of *CrossSOE_Dum* (*CrossSOE_Num*) and *CrossFOR_Dum* (*CrossFOR_Num*) are significantly negative, while the coefficient of *CrossNSOE_Dum* (*CrossNSOE_Num*) is insignificant. These results suggest that common state ownership and common foreign ownership are likely to act as more active monitors and play a stronger governance role. These findings are consistent with previous studies, which show that state ownership enhances firm performance through effective monitoring (Chen et al., 2009), and that foreign investors have superior experience in information management than domestic investors (Gul et al., 2010).

5. Conclusion

In this paper, we empirically examine whether and how common ownership impacts corporate violations using a sample of Chinese A-share listed firms from 2007 to 2022. Consistent with the view of the "synergistic governance" effect, we find significant evidence of a negative relationship between common ownership and firms' violation behaviors. This negative relationship remains robust after addressing endogeneity concerns through various methodologies. We also conduct additional robustness tests, including alternative independent variables, shortened sample periods, alternative models, and alternative fixed effects. We find that the potential mechanisms are improved corporate governance and enhanced information transparency. The negative relationship between common ownership and corporate violations is more pronounced among firms with a higher external monitoring and those with a lower proportion of state-owned shares. By distinguishing between different types of common ownership, we identify that common state ownership and common foreign ownership serve as key forces in curbing corporate violations. Overall, we shed light on the role of common ownership in deterring a firm's violation behaviors. Thus, this study contributes to the literature on the determinants of corporate violations and also enriches the existing literature related to common ownership.

These findings provide policy implications for governments and corporations. In terms of the public interest, the Chinese government should take advantage of the positive role of common ownership to stabilize the capital market. This stabilization can be achieved by encouraging practices that enhance transparency and accountability among firms. In terms of business, commercial enterprises should proactively introduce and actively manage common ownership to ensure their "synergistic governance", which helps mitigate the risk of corporate violations and enhances long-term value. In addition, many countries are strengthening the regulatory requirements surrounding common ownership; however, they face significant debate about whether common ownership is beneficial or detrimental. Hence, the insights from this study may be applicable to countries far beyond China, helping to inform policy and regulatory frameworks in broader international contexts.

CRediT authorship contribution statement

Rongrong Yao: Writing – original draft, Visualization, Software, Resources, Data curation, Conceptualization. **Min Xiao:** Conceptualization, Writing – review & editing.

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Appendix A. Appendix

Appendix Table 1
Variable definitions.

Variable	Definitions
Dependent variable	
VioDummy	A dummy variable that equals 1 when a firm is sanctioned for violations by regulatory authorities, and 0 otherwise.
VioNum	The sum of violations of the firm.
Independent variable	
Cross5_Dum	A dummy variable that equals 1 if the focal firm and at least one of its industry peers are simultaneously held by at least one blockholder in any quarter of the fiscal year, and 0 otherwise.

(*continued on next page*)

[3] The regression results reported in Panel B of Table 7 do not control for *CrossInst_Dum* (*CrossInst_Num*) and *CrossIncDNum* (*CrossInd_Num*). The results remain unchanged when these two variables are controlled.

Appendix Table 1 (*continued*)

Variable	Definitions
Cross5_Num	Measured by the natural logarithm of 1 plus the average number of common blockholders that hold the focal firm and industry peers across the four quarters of a fiscal year.
Control variables	
Size	The natural logarithm of total assets.
Lev	Total liabilities divided by total assets.
ROA	Net income divided by total assets.
OCF	Net cash flow from operating activities scaled by total assets.
Growth	The change in sales from year t-1 to year t, divided by sales in year t-1.
Invest	Capital expenditure scaled by total assets.
Top1	The shareholdings of the largest shareholder.
Indirect	The ratio of independent directors to total directors.
Dual	A dummy variable that equals to 1 when the firm's CEO is also the chairman, and 0 otherwise.
SOE	A dummy variable that equals 1 if the firm is controlled by the state, and 0 otherwise.
Fix	The ratio of tangible fixed assets to total assets.
Age	The natural logarithm of one plus the number of years since the firm was listed.
BIG4	A dummy variable equals one if the firm is audited by Big 4 auditors, and zero otherwise.
Other variables	
CSI300in	A dummy variable that equals 1 if a firm newly adds to the CSI 300 index in that year, and 0 otherwise.
Dividend	A dummy variable that equals 1 if the firm pays a cash dividend, and 0 otherwise.
Density	The density of the ownership network is calculated by multiplying the stakes held by each common shareholder in the two firms and then aggregating these products across all common shareholders within the same industry. See Formula (3).
Cross5_Share	The average shareholdings of common ownership that holds the focal firm and industry peers across the four quarters of a fiscal year.
Cross3_Dum	A dummy variable that equals 1 if the focal firm and at least one of its industry peers are simultaneously held by at least one blockholder who own at least 3 % of a firm's outstanding shares in any quarter of the fiscal year, and 0 otherwise.
Cross1_Dum	A dummy variable that equals 1 if the focal firm and at least one of its industry peers are simultaneously held by at least one blockholder who own at least 1 % of a firm's outstanding shares in any quarter of the fiscal year, and 0 otherwise.
CODirector	The ratio of directors appointed by common investors to total directors.
ICQ	The Internal Control Index was published by Diebold to present the quality of a firm's internal control.
SGA	The ratio of the amount of selling, general, and administrative expenses to total sales.
ORR	The ratio of other receivables to total assets.
REM	The real earnings management follows Roychowdhury (2006) and is specified as *REM* = standardized *R_Prod* - standardized *R_Cfo* - standardized *R_Disexp*. *R_Prod* represents abnormal production costs. *R_Cfo* represents abnormal cash flows from operations. *R_Disexp* represents abnormal discretionary expenses. Then we take absolute values as a proxy for real earnings management.
ASY	The index of information asymmetry is the first principal component of *LR*, *ILL*, and *GAM* following Amihud et al. (1997), Amihud (2002), and Pástor and Stambaugh (2003). Amihud et al. (1997) construct liquidity ratio: $LR_{i,t} = -\frac{1}{D_{i,t}}\sum_{k=1}^{D_{i,t}}\sqrt{\frac{V_{i,t}(k)}{\lvert r_{i,t}(k)\rvert}}$. Amihud (2002) constructs illiquidity ratio: $ILL_{i,t} = \frac{1}{D_{i,t}}\sum_{k=1}^{D_{i,t}}\sqrt{\frac{\lvert r_{i,t}(k)\rvert}{V_{i,t}(k)}}$. $r_{i,t}(k)$ denotes the stock return of firm i on the kth trading day of year t. $V_{i,t}(k)$ denotes daily volume, and $D_{i,t}$ denotes the number of trading days in the year. Pástor and Stambaugh (2003) propose the return reversal (*GAM*) to measure of liquidity.
Media	The natural logarithm of the number of online financial negative news articles plus 1.
SOE_Share	The shareholdings of the firm's proportion of state ownership.
CrossInst_Dum	A dummy variable that equals 1 if the focal firm and at least one of its industry peers are simultaneously held by at least one common institutional ownership in any quarter of the fiscal year, and 0 otherwise.
CrossInst_Num	The natural logarithm of 1 plus the average number of common institutional ownership that hold the focal firm and industry peers across the four quarters of a fiscal year.
CrossEnt_Dum	A dummy variable that equals 1 if the focal firm and at least one of its industry peers are simultaneously held by at least one common enterprise ownership in any quarter of the fiscal year, and 0 otherwise.
CrossEnt_Num	The natural logarithm of 1 plus the average number of common enterprise ownership that hold the focal firm and industry peers across the four quarters of a fiscal year.
CrossInd_Dum	A dummy variable that equals 1 if the focal firm and at least one of its industry peers are simultaneously held by at least one common individual ownership in any quarter of the fiscal year, and 0 otherwise.
CrossInd_Num	The natural logarithm of 1 plus the average number of common individual ownership that hold the focal firm and industry peers across the four quarters of a fiscal year.
CrossSOE_Dum	A dummy variable that equals 1 if the focal firm and at least one of its industry peers are simultaneously held by at least one common state ownership in any quarter of the fiscal year, and 0 otherwise.
CrossSOE_Num	The natural logarithm of 1 plus the average number of common state ownership that hold the focal firm and industry peers across the four quarters of a fiscal year.
CrossNSOE_Dum	A dummy variable that equals 1 if the focal firm and at least one of its industry peers are simultaneously held by at least one common non-state ownership in any quarter of the fiscal year, and 0 otherwise.
CrossNSOE_Num	The natural logarithm of 1 plus the average number of common non-state ownership that hold the focal firm and industry peers across the four quarters of a fiscal year.
CrossFOR_Dum	A dummy variable that equals 1 if the focal firm and at least one of its industry peers are simultaneously held by at least one common foreign ownership in any quarter of the fiscal year, and 0 otherwise.
CrossFOR_Num	The natural logarithm of 1 plus the average number of common foreign ownership that hold the focal firm and industry peers across the four quarters of a fiscal year.

Appendix B. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.pacfin.2025.102926.

References

Aggarwal, R., Erel, I., Ferreira, M., Matos, P., 2011. Does governance travel around the world? Evidence from institutional investors. J. Financ. Econ. 100 (1), 154–181.
Agrawal, A., Chadha, S., 2005. Corporate governance and accounting scandals. J. Law Econ. 48 (2), 371–406.
Amihud, Y., 2002. Illiquidity and stock returns: cross-section and time-series effects. J. Financ. Mark. 5 (1), 31–56.
Amihud, Y., Mendelson, H., Lauterbach, B., 1997. Market microstructure and securities values: evidence from the Tel Aviv Stock exchange. J. Financ. Econ. 45 (3), 365–390.
Antón, M., Ederer, F., Giné, M., Schmalz, M., 2023. Common ownership, competition, and top management incentives. J. Polit. Econ. 131 (5), 1294–1355.
Athira, A., Lukose, P.J.J., 2023. Do common institutional owners' activisms deter tax avoidance? Evidence from an emerging economy. Pac. Basin Financ. J. 80, 102090.
Azar, J., Schmalz, M.C., Tecu, I., 2018. Anticompetitive effects of common ownership. J. Financ. 73 (4), 1513–1565.
Banerjee, P., Dhole, S., Mishra, S., 2023. Operating performance during the COVID-19 pandemic: is there a business group advantage? Pac. Basin Financ. J. 79, 102031.
Bell, T.B., Carcello, J.V., 2000. A decision aid for assessing the likelihood of fraudulent financial reporting. Audi. A J. Pract. Theory 19 (1), 169–184.
Bleck, A., Liu, X., 2007. Market transparency and the accounting regime. J. Account. Res. 45 (2), 229–256.
Brooks, C., Chen, Z., Zeng, Y., 2018. Institutional cross-ownership and corporate strategy: the case of mergers and acquisitions. J. Corp. Finan. 48, 187–216.
Chen, G., Firth, M., Gao, D.N., Rui, O.M., 2006. Ownership structure, corporate governance, and fraud: evidence from China. J. Corp. Finan. 12 (3), 424–448.
Chen, G., Firth, M., Xu, L., 2009. Does the type of ownership control matter? Evidence from China's listed companies. J. Bank. Financ. 33 (1), 171–181.
Chen, S., Ma, H., Wu, Q., Zhang, H., 2023a. Common institutional ownership and stock price crash risk. Contemp. Account. Res. 41 (1), 679–711.
Chen, Y., Wu, S., Zhou, Y., Huo, D., 2023b. Gambling culture and corporate violations: evidence from China. Pac. Basin Financ. J. 80, 102099.
Cheng, W., Li, C., Zhao, T., 2024. The stages of enterprise digital transformation and its impact on internal control: evidence from China. Int. Rev. Financ. Anal. 92, 103079.
Chetty, R., Looney, A., Kroft, K., 2009. Salience and taxation: theory and evidence. Am. Econ. Rev. 99 (4), 1145–1177.
Chhaochharia, V., Grinstein, Y., Grullon, G., Michaely, R., 2017. Product market competition and internal governance: Evidence from the Sarbanes–Oxley Act. Manag. Sci. 63 (5), 1405–1424.
Cragg, J.G., Donald, S.G., 1993. Testing identifiability and specification in instrumental variable models. Economet. Theor. 9 (2), 222–240.
Dai, J., Xu, R., Zhu, T., Lu, C., 2024. Common institutional ownership and opportunistic insider selling: evidence from China. Pac. Basin Financ. J. 88, 102580.
Donelson, D.C., Ege, M.S., McInnis, J.M., 2017. Internal control weaknesses and financial reporting fraud. Audit. J. Pract. Theory 36 (3), 45–69.
Dyck, A., Volchkova, N., Zingales, L., 2008. The corporate governance role of the media: evidence from Russia. J. Financ. 63 (3), 1093–1135.
Dyck, A., Morse, A., Zingales, L., 2010. Who blows the whistle on corporate fraud? J. Financ. 65 (6), 2213–2253.
Edmans, A., Levit, D., Reilly, D., 2019. Governance under common ownership. Rev. Financ. Stud. 32 (7), 2673–2719.
Efendi, J., Srivastava, A., Swanson, E.P., 2007. Why do corporate managers misstate financial statements? The role of option compensation and other factors. J. Financ. Econ. 85 (3), 667–708.
Ferrara, E.L., Chong, A., Duryea, S., 2012. Soap operas and fertility: evidence from Brazil. Am. Econ. J. Appl. Econ. 4 (4), 1–31.
Ferreira, M.A., Matos, P., 2008. The colors of investors' money: the role of institutional investors around the world. J. Financ. Econ. 88 (3), 499–533.
Fich, E.M., Shivdasani, A., 2007. Financial fraud, director reputation, and shareholder wealth. J. Financ. Econ. 86 (2), 306–336.
Gul, F.A., Kim, J.-B., Qiu, A.A., 2010. Ownership concentration, foreign shareholding, audit quality, and stock price synchronicity: evidence from China. J. Financ. Econ. 95 (3), 425–442.
Hansen, R.G., Lott, J.R., 1996. Externalities and corporate objectives in a world with diversified shareholder/consumers. J. Finan. Quantitat. Anal. 31 (1), 43–68.
Hart, O.D., 1983. The market mechanism as an incentive scheme. Bell J. Econ. 14 (2), 366–382.
He, J., Huang, J., 2017. Product market competition in a world of cross-ownership: evidence from institutional blockholdings. Rev. Financ. Stud. 30 (8), 2674–2718.
He, J., Huang, J., Zhao, S., 2019. Internalizing governance externalities: the role of institutional cross-ownership. J. Financ. Econ. 134 (2), 400–418.
Heckman, J.J., 1979. Sample selection bias as a specification error. Econometrica 47 (1), 153–161.
Heese, J., Pérez-Cavazos, G., Peter, C.D., 2022. When the local newspaper leaves town: the effects of local newspaper closures on corporate misconduct. J. Financ. Econ. 145 (2), 445–463.
Holzman, E.R., Miller, B.P., Williams, B.M., 2021. The local spillover effect of corporate accounting misconduct: evidence from city crime rates. Contemp. Account. Res. 38 (3), 1542–1580.
Hou, W., Kuo, J.-M., Lee, E., 2012. The impact of state ownership on share price informativeness: the case of the Split share structure reform in China. Br. Account. Rev. 44 (4), 248–261.
Hsu, C., Ma, Z., Zhou, K., 2023. Does Common Ownership Affect Employee Welfare? Evidence from corporate pension funding. SSRN Electronic Journal. https://doi.org/10.2139/ssrn.4388014.
Jang, I.J., Kang, N., Yezegel, A., 2022. Common ownership, price informativeness, and corporate investment. J. Bank. Financ. 135, 106373.
Jiang, G., Lee, C.M.C., Yue, H., 2010. Tunneling through intercorporate loans: the China experience. J. Financ. Econ. 98 (1), 1–20.
Jiang, Z., Liu, B., Liu, J., Ying, Q., 2022. Media attention and regulatory efficiency of corporate violations: evidence from China. J. Account. Public Policy 41 (3), 106931.
Jin, L., Myers, S., 2006. R^2 around the world: new theory and new tests. J. Financ. Econ. 79 (2), 257–292.
Kang, J.-K., Luo, J., Na, H.S., 2018. Are institutional investors with multiple blockholdings effective monitors? J. Financ. Econ. 128 (3), 576–602.
Karpoff, J.M., Lou, X., 2010. Short sellers and financial misconduct. J. Financ. 65 (5), 1879–1913.
Kedia, S., Rajgopal, S., 2011. Do the SEC's enforcement preferences affect corporate misconduct? J. Account. Econ. 51 (3), 259–278.
Khanna, V., Kim, E., Lu, Y., 2015. CEO connectedness and corporate fraud. J. Financ. 70 (3), 1203–1252.
La Porta, R., Lopez-de-Silanes, F., Shleifer, A., Vishny, R., 2002. Investor protection and corporate valuation. J. Financ. 57 (3), 1147–1170.
Lennox, C.S., Francis, J.R., Wang, Z., 2012. Selection models in accounting research. Account. Rev. 87 (2), 589–616.
Lewellen, J., Lewellen, K., 2022. Institutional investors and corporate governance: the incentive to be engaged. J. Financ. 77 (1), 213–264.
Lewellen, K., Lowry, M., 2021. Does common ownership really increase firm coordination? J. Financ. Econ. 141 (1), 322–344.
Li, K., Liu, X., Mai, F., Zhang, T., 2021. The role of corporate culture in bad times: evidence from the COVID-19 pandemic. J. Financ. Quant. Anal. 56 (7), 2545–2583.
Lin, L., Xiao, M., Yao, R., Zhang, X., 2024. Product market liberalization and corporate cash holdings: evidence from China. Pac. Basin Financ. J. 86, 102468.
Liu, C., He, J., 2024. Does common institutional ownership improve the similarity of behavior between firms? Evidence from China. Pac. Basin Financ. J. 83, 102225.
Liu, X., Boubaker, S., Liao, J., Yao, S., 2024. The rise of common state ownership and corporate environmental performance. Br. Account. Rev. 101368.
Lu, R., Shen, Q., Wang, T., Zhang, X., 2022. Frenemies: corporate advertising under common ownership. Manag. Sci. 68 (6), 4645–4669.
Miller, G.S., 2006. The press as a watchdog for accounting fraud. J. Account. Res. 44 (5), 1001–1033.

Park, J., Sani, J., Shroff, N., White, H., 2019. Disclosure incentives when competing firms have common ownership. J. Account. Econ. 67 (2–3), 387–415.
Pástor, Ľ., Stambaugh, R.F., 2003. Liquidity risk and expected stock returns. J. Polit. Econ. 111 (3), 642–685.
Pawliczek, A., Skinner, A.N., Zechman, S.L.C., 2022. Facilitating tacit collusion through voluntary disclosure: evidence from common ownership. J. Account. Res. 60 (5), 1651–1693.
Ramalingegowda, S., Utke, S., Yu, Y., 2021. Common institutional ownership and earnings management. Contemp. Account. Res. 38 (1), 208–241.
Roychowdhury, S., 2006. Earnings management through real activities manipulation. J. Account. Econ. 42 (3), 335–370.
Shan, Y.G., 2013. Can internal governance mechanisms prevent asset appropriation? Examination of type I tunneling in China. Corp. Gov. 21 (3), 225–241.
Shehzad, K., Liu, X., Kazouz, H., 2020. COVID-19's disasters are perilous than global financial crisis: a rumor or fact? Financ. Res. Lett. 36, 101669.
Singh, M., Davidson III, W.N., 2003. Agency costs, ownership structure and corporate governance mechanisms. J. Bank. Financ. 27 (5), 793–816.
Stock, J.H., Yogo, M., 2005. Testing for weak instruments in Linear IV regression. Nber Techn. Work. Papers. 14 (1), 80–108.
Wang, T.Y., 2013. Corporate securities fraud: insights from a new empirical framework. J. Law Econ. Org. 29 (3), 535–568.
Wang, X., Xu, L.C., Zhu, T., 2004. State-owned enterprises going public: the case of China. Econ. Transit. 12 (3), 467–487.
Wang, T.Y., Winton, A., Yu, X., 2010. Corporate fraud and business conditions: evidence from IPOs. J. Financ. 65 (6), 2255–2292.
Wu, J., Zhan, X., Xu, H., Ma, C., 2023. The economic impacts of COVID-19 and city lockdown: early evidence from China. Struct. Chang. Econ. Dyn. 65, 151–165.
Xu, Z., Wang, S., Ye, J., 2024. The effect of digitization on corporate fraud detection evidence from China. Int. Rev. Financ. Anal. 96, 103788.
Yao, S., Guo, X., Sensoy, A., Goodell, J.W., Cheng, F., 2024. Collusion or governance? Common ownership and corporate risk-taking. Corp. Gov. 32 (4), 645–669.
Zhang, J., 2018. Public governance and corporate fraud: evidence from the recent anti-corruption campaign in China. J. Bus. Ethics 148 (2), 375–396.